TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Electrospun Nanofiber Coatings for High-Efficiency and Durable Storage of Liquified Hydrogen			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), **Hydrogen (√),** Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	Hydrogen	Polymeric Materials	Materials for Energy/Environmental Industries
Concept of Proposed Technology	The proposed technology is a nanofiber-based coating platform designed to improve thermal insulation, durability, and operational reliability of cryogen (e.g., liquid hydrogen) storage. This technology leverages electrospinning-based manufacturing to create highly porous, hierarchical fibrous coatings with tailored thermal, optical, and mechanical properties. Originally developed for cryogenic storage in space [1], this technology can also be useful for terrestrial hydrogen storage and transport. The core concept is to use engineered nanofiber architectures (Figure 1-Left) to minimize parasitic heat transfer into hydrogen storage vessels while maintaining low weight and long-term environmental stability. By controlling fiber diameter, porosity, surface chemistry, and coating thickness, the material properties can be tuned [2] to suppress conductive and radiative heat transfer while remaining compatible with industrial coating processes. Its potential uses could include: • Cryogenic and compressed liquified hydrogen storage tanks			

- Coatings for hydrogen transport vessels and piping systems
- Insulation layers for hydrogen refueling infrastructure
- Thermal protection for hydrogen-based mobility and energy systems
- Due to large solar reflectance of these materials, thermal loads on buildings can also be reduced [3]

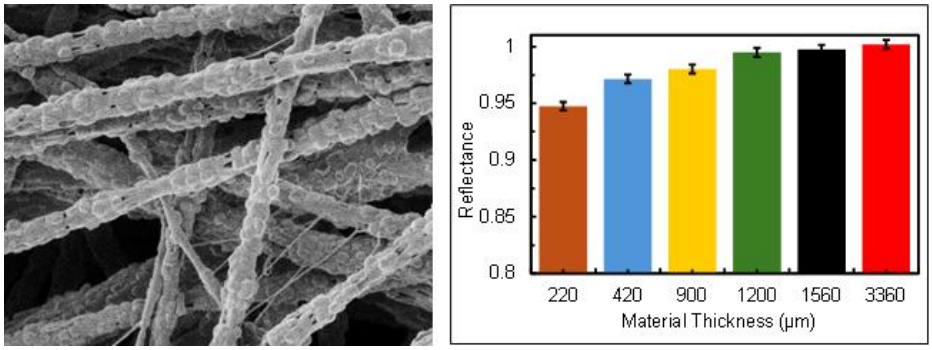

Figure 1. (Left) Electrospun PTFE-PEO Fibers. (Right) Average solar reflectance versus PTFE:PEO material thickness

The proposed technology is differentiated from conventional coatings by its microstructure-driven performance rather than reliance on the material property alone. Unlike traditional (multi-layer) insulation materials that use thin metallized polymer sheets, this approach leverages light scattering by porous materials, decoupling solar reflectance, thermal radiative exchange, and conductive heat transfer processes through a layered configuration. An electrospun nanofiber outer coating layer provides exceptionally high solar reflectance (Figure 1-Right) and environmental durability [4], while radiative heat transfer will be suppressed using low-emissivity internal surfaces along with vacuum or low-conductivity separation layer optimized for terrestrial conditions. This modular architecture enables the same materials platform to be reconfigured for distinct operating environments, such as land-based hydrogen infrastructure and space systems, by adjusting layer placement and optical function rather than reformulating the material itself. As a result, this technology offers a unique combination of performance tunability, manufacturability, and application versatility that is not achievable with conventional insulation solutions, making it well suited for translation through joint R&D with industrial partners.

Rationale for Joint R&D Project

While the core innovation, materials concepts and laboratory-scale performance have been established at RPI [5], successful translation to real hydrogen storage systems requires close collaboration with industry partners, including small and medium-sized enterprises with expertise in manufacturing, coating integration, and system-level validation.

A joint R&D project is essential to:

- Adapt nanofiber coatings to industry-standard tank materials and geometries
- Address manufacturability, coating uniformity, and scalability
- Evaluate long-term durability under thermal cycling and environmental exposure
- Validate performance under realistic hydrogen storage operating conditions

Korean SMEs are particularly well positioned to contribute expertise in advanced materials processing, hydrogen system components, and industrial qualification testing. The collaboration will enable accelerated translation from laboratory prototypes (TRL 3) to application-ready coating systems (TRL 5) suitable for planning commercialization under the TIPA International Joint R&D Program.

Key Words

1. Electrospun nanofiber coatings
2. Liquid hydrogen storage
3. Thermal insulation materials
4. Nanofibrous solar reflectors
5. Thermal insulation for cryogenic storage

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	The primary objective of this project is to validate integrated thermal insulation architectures for terrestrial hydrogen storage systems that leverage nanofiber-based coatings as an outer environmental barrier, combined with low-emissivity internal surfaces and thermal separation layers optimized for terrestrial operating environments. Specific objectives include: • Scaled-up manufacturing of nanofiber-based outer coatings that provide exceptionally high solar reflectance, minimizing solar-driven heat gain in outdoor environments • Engineer system-level insulation configurations in which the nanofiber layer is paired with low-emissivity internal surfaces to suppress radiative heat transfer from the ambient • Evaluate vacuum gaps or low-conductivity porous interlayers as thermal separators when vacuum insulation is impractical • Optimize coating thickness, architecture, and placement (via modeling) to balance thermal performance, durability, and manufacturability • Demonstrate thermal performance improvements at the component or subsystem level, relevant to stationary hydrogen storage infrastructure This approach ensures that the proposed technology is optimized for land-based hydrogen storage, where ambient temperatures are near 300 K and radiative heat exchange must be carefully managed.			
R&D Target Specifications	The target specifications for this project emphasize system-level thermal performance, rather than material properties in isolation. **Solar Heat Gain Reduction:** Achieve very high solar reflectance (>95%, with targets approaching >99%) for the outer nanofiber layer, significantly reducing direct solar heat absorption under outdoor exposure. **Radiative Heat Transfer Control: (a)** Use low-emissivity internal surfaces (e.g., aluminum or coated metallic substrates) to suppress radiative heat transfer from the ambient environment at ~300 K. (b) Demonstrate that the combined system exhibits net reduction in radiative heat gain relative to conventional insulation configurations. **Thermal Separation Layer Performance: (a)** Validate vacuum gaps or low-thermal-conductivity porous interlayers that reduce conductive and radiative coupling between the outer coating and the hydrogen storage vessel. (b) Achieve effective thermal resistance improvements without requiring large insulation thickness or complex structural modifications. **Durability and Environmental Stability:** Maintain solar reflectance and structural integrity of the outer nanofiber layer under prolonged UV exposure, humidity, and thermal cycling typical of terrestrial environments. **Manufacturability and Integration:** Demonstrate compatibility with existing hydrogen storage tank materials and geometries, enabling retrofit or modular integration without major system redesign.			
R&D Plan	**Phase 1 (Months 0–6): RPI-Industry Collaboration** • Hold periodic meetings to finalize and confirm projects goals and R&R with industrial partner • Optimize nanofiber outer coatings for solar reflectance and environmental durability • Identify suitable low-emissivity internal surface treatments **Phase 2.1 (Months 7–12): Architecture Design and Material Optimization** • Model radiative–conductive heat transfer in multi-layer insulation configurations			

	Phase 2.2 (Months 13–24): Integrated Insulation Development • Fabricate layered insulation assemblies (outer nanofiber layer + thermal separation + reflective substrate) • Evaluate vacuum and non-vacuum thermal separation strategies • Perform laboratory-scale thermal testing under terrestrial ambient conditions **Phase 2.3 (Months 25–36): Subsystem Validation** • Validate thermal performance on representative hydrogen storage components • Conduct durability testing under outdoor-relevant environmental cycling • Generate design guidelines and performance data for industrial deployment

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	RPI will lead the **fundamental design, modeling, and experimental validation** of the proposed integrated insulation architecture, including: • Design and fabrication of nanofiber-based outer coatings with tailored optical properties • Radiative–conductive heat transfer modeling of multi-layer insulation systems under terrestrial conditions • Experimental characterization of thermal performance, durability, and environmental stability • Development of design rules linking material architecture to system-level thermal performance • Knowledge transfer and technical coordination with Korean partners RPI will provide laboratory infrastructure, modeling expertise, and materials characterization capabilities necessary to establish performance baselines and guide translation.
Expected R&R of Korean SME Partner	The Korean SME partner will lead **engineering translation, manufacturing adaptation, and application validation**, including: • Providing hydrogen storage system requirements and operating constraints • Integrating the proposed insulation architecture into representative storage systems • Conducting application-relevant testing and validation • Supporting scaled-up manufacturing, cost analysis, and commercialization planning The SME partner's involvement ensures that the developed technology is **deployable within real hydrogen infrastructure**, rather than remaining a laboratory-scale material concept.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project will deliver a **versatile nanofiber-based thermal management technology platform** that can be configured to address both **space** and **terrestrial hydrogen storage applications**, with architecture-specific optimization tailored to the distinct thermal environments of each domain. **Impact on Terrestrial Hydrogen Storage and Infrastructure:** For land-based hydrogen storage systems operating in ambient environments near 300 K, the project will deliver **integrated insulation architectures** that prioritize reduction of solar-driven heat gain and suppression of net radiative heat transfer from the surroundings. Key terrestrial impacts include: • **Substantial reduction in solar heat absorption** through ultra-high solar reflectance nanofiber outer layers, leading to lower thermal loads on hydrogen storage vessels • **Net reduction in radiative heat gain** by combining the nanofiber outer coating with low-

emissivity internal surfaces and thermal separation layers (vacuum gaps or low-conductivity porous insulation)

- **Improved thermal efficiency and reduced boil-off losses**, particularly for outdoor storage tanks and infrastructure exposed to diurnal and seasonal solar variations
- **Enhanced durability and service lifetime**, enabled by environmentally resilient nanofiber coatings that withstand UV exposure, humidity, and thermal cycling

These advances directly support **lower operational energy consumption, reduced maintenance requirements, and improved reliability** of hydrogen storage infrastructure. The system-level design approach also enables **retrofit compatibility** with existing tanks and components, facilitating adoption by Korean SMEs without requiring major redesign of current systems.

Impact on Space and Extraterrestrial Applications:

In space and extraterrestrial environments, where the effective background temperature is extremely low, the same nanofiber materials provide additional benefits through **simultaneous high solar reflectance and strong thermal emission**. Key space-specific impacts include:

- **Efficient rejection of absorbed heat through thermal radiation**, enabling passive temperature regulation of spacecraft surfaces and cryogenic propellant tanks
- **Reduced dependence on active thermal control systems**, freeing power and mass for mission-critical functions
- **Demonstrated resistance to harsh space environments**, including UV radiation and atomic oxygen exposure, supporting long-duration missions

The ability to tailor nanofiber composition, architecture, and system placement allows the technology to be optimized for **diverse mission profiles**, including spacecraft thermal control and long-term cryogenic fluid storage in space.

Broader Industrial and Economic Impact

By establishing a **single materials platform with configurable system-level implementations**, this project enables broad applicability across multiple sectors while maintaining a focused pathway to commercialization. Broader impacts include:

- Accelerated **international collaboration** between RPI and Korean SMEs in hydrogen and advanced materials technologies
- Development of **scalable manufacturing pathways** suitable for SME adoption
- Strengthening Korea's competitiveness in hydrogen infrastructure, advanced insulation materials, and energy systems
- Creation of follow-on opportunities under the TIPA International Joint R&D framework and beyond

Collectively, the outcomes of this project will advance both **near-term terrestrial hydrogen infrastructure** and **long-term space exploration technologies**, while delivering practical, industry-ready solutions grounded in sound thermal physics.

6. References

[1] https://www.nasa.gov/directorates/stmd/space-tech-research-grants/three-dimensional-hierarchical-structures-as-multi-layer-insulation-for-terrestrial-and-space-applications/

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[3] Ibekwe, C, Wang, X, Hartwig, JW, Swanger, AM, & Narayan, S. "Manufacturing, Characterization and Terrestrial Passive Radiative Cooling Performance of Nanofibrous PTFE-PEO Composite Thermal Coatings." *Proceedings of the ASME 2024 International Mechanical Engineering Congress and Exposition*. *Volume 9: Heat Transfer and Thermal Engineering*. Portland, Oregon, USA. November 17–21, 2024. V009T11A029. ASME. https://doi.org/10.1115/IMECE2024-146008

[4] Ibekwe, C, Wang, X, Hartwig, JW, Feller, JR, Swanger, AM, & Narayan, S. "Electrospun Silica Nanofibers as Passive Thermal Control Materials for the Extreme Environment of Low Earth Orbit." *Proceedings of the ASME 2024 International Mechanical Engineering Congress and Exposition*. *Volume 9: Heat Transfer and Thermal Engineering*. Portland, Oregon, USA. November 17–21, 2024. V009T11A037. ASME. https://doi.org/10.1115/IMECE2024-144206

[5] https://patents.google.com/patent/US20250320634A1/en

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization